

Ella, Francis Agbam¹, Normala Halimoon^{1}, Ferdaus Mohamat Yusuff¹, Wan Lutfi Wan Johari¹, Abdulwakil Olawale Saba², Ella, Adakole Boniface³, Effiong, Mmandu Uwem⁴*

¹Department of Environmental Science and Technology, Faculty of Forestry and Environment, Universiti Putra Malaysia, 43300 Serdang, Selangor, Malaysia.

²Department of Fisheries, Faculty of Science, Lagos State University, Ojo, Lagos, Nigeria

³Affiliation: Department of Environment Science and Management, Faculty of Science, National Open University of Nigeria.

⁴Department of Animal and Environmental Biology, University of Uyo, Uyo, Akwa-Ibom, State, Nigeria

****Corresponding author:** mala_upm@upm.edu.my*

Abstract

Conventional trace element pollution-index studies characterize chemical contamination but rarely test whether it structures biological communities, leaving risk classifications unvalidated. This study integrates trace element, physicochemical, and macroinvertebrate data across five Kuyoh River stations (Control, S1–S4), sampled monthly across dry and wet seasons, three replicates per station per visit (physicochemical $n = 24$ per station; trace elements $n = 12$ per station, dry season). Separating raw variables from indices avoided circular correlation. Physicochemical PCA ($KMO = 0.844$, Bartlett's $\chi^2(15) = 677.75$, $p < 0.001$) resolved an organic pollution gradient (PC1: 69.9%; NH_3 -N, DO, BOD_5) and a pH-associated axis (PC2: 16.3%). Trace element PCA ($KMO = 0.656$, Bartlett's $\chi^2(15) = 153.36$, $p < 0.001$) resolved an As–Zn–Mn gradient (PC1: 53.0%) and a Cd–Cu point-source factor (PC2: 16.8%). HPI, ERI, and MPI rose monotonically from Control to S3/S4; ERI classified S3/S4 as very high risk (1099, 1189), while HPI stayed below its critical value of 100 at every station (2.11–9.58). CCA of dry-season data linked family abundance to the twelve-variable dataset, confirming organic enrichment as the dominant gradient (CCA1: 89.3%), while cadmium and copper loaded independently on a secondary axis (CCA2: 9.2%), orthogonal to organic pollution. Trace element contamination thus produced a biological structuring effect distinguishable from organic stress, providing validated evidence of independent point-source effects and a transferable framework for source apportionment in tropical urban catchments.

Keywords : Cadmium and Copper; Macroinvertebrate Assemblage; Organic Enrichment; Tropical Urban River.

1-Introduction

Trace element contamination of urban rivers is well-documented, with tools such as the Heavy Metal Pollution Index (HPI), Metal Pollution Index (MPI), Ecological Risk Index (ERI), and Principal Component Analysis (PCA) widely applied (Ahmad et al., 2026; Castro-Elenes et al., 2021; Dheeraj et al., 2024). Studies in Malaysia, Vietnam, and China show elevated downstream contamination at industrial stations versus upstream references (Bian et al., 2016; Hoang et al., 2024; Bahri et al., 2024), yet often cannot determine whether contamination independently structures biological communities (Let et al., 2022; Mohd Najib et al., 2025).

Two gaps motivate this study, one interpretive and one methodological, requiring chemical and biological response be treated as independent evidence, not substituted by indices (Moulinec et al., 2025). Indices quantify exceedance against thresholds but do not confirm measurable community effects (Cuss et al., 2020; Posthuma et al., 2020) and may be classified without testing organism response (Friberg, 2010; Hering et al., 2006). High ERI values may occur without compositional response, while moderate values may coincide with taxa restructuring, especially for Cd and Cu (Clements et al., 2000; Clements & Rohr, 2009; Liang et al., 2024).

This framework is fragile in tropical urban rivers for three reasons: First, indices carry toxicological weightings from other systems: the ERI weights cadmium thirty times more than zinc or manganese, derived from temperate lake sediments (Hakanson, 1980), causing high-factor elements to dominate scores regardless of impact, a weakness K. R. Singh et al. (2020) identify, addressed via integrated biological terms (Jaiswal & Pandey, 2019). Second, dissolved-phase concentrations in monsoonal catchments are strongly seasonal, so a single-sample index may reflect hydrology, not exposure (Pakoksung et al., 2025). Third, urban rivers are multi-stressor systems where trace elements co-occur with organic enrichment and hypoxia along collinear gradients; chemical pollution constrains ecological status, but its effect is confounded with hydromorphological/nutrient pressures (Feld et al., 2014; Johnson & Hering, 2009; Posthuma et al., 2020).

Bioassessment faces a mirror limitation: Clements et al. (2000) showed heavy metals structure benthic communities independently of nutrients generalized by Clements & Rohr, (2009). Let et al. (2022) documented EPT loss under combined metal-wastewater exposure while Moulinec et al. (2025) confirmed biomonitoring detects pressure chemical monitoring classifies acceptable. Cuss et al. (2020) showed speciation, not total concentration, governs relevance. Cd disrupts calcium transport and

displaces Zn at enzyme sites independent of oxygen depletion, predicting Cd and Cu structure assemblages orthogonal to organic enrichment.

Most evidence comes from mining-influenced temperate streams without sewage, and whether this separation persists in tropical rivers remains untested. CCA offers a methodological bridge, yet its combination with pollution indices remains rare in Southeast Asia, where studies focus on sediment-phase links (Bian et al., 2016; Jumaat & Hamid, 2023) rather than dissolved-phase concentrations structuring assemblages independently of organic gradients. Liang et al. (2024) and Zhang et al. (2023) linked assemblages to sediment metals or bioaccumulation without assemblage-level testing, while chemical-only studies (Sama et al., 2024; Mazilamani et al., 2025) leave Cd and Cu's contribution unresolved.

A second limitation is weak statistical robustness: studies report correlation/PCA results from station-mean values without replicates or seasonal support (Singh et al., 2024; Snousy et al., 2021), and composite indices can inflate PCA variance via circularity. This study addresses both identified gaps with an integrated framework, directly linking replicate-based trace element and physicochemical data to macroinvertebrate composition through CCA, distinguishing trace-element from organic-pollution signals, with raw-variable/index separation ensuring robust, transparent HPI, ERI, and MPI reporting (Haque et al., 2022; Jaiswal & Pandey, 2019).

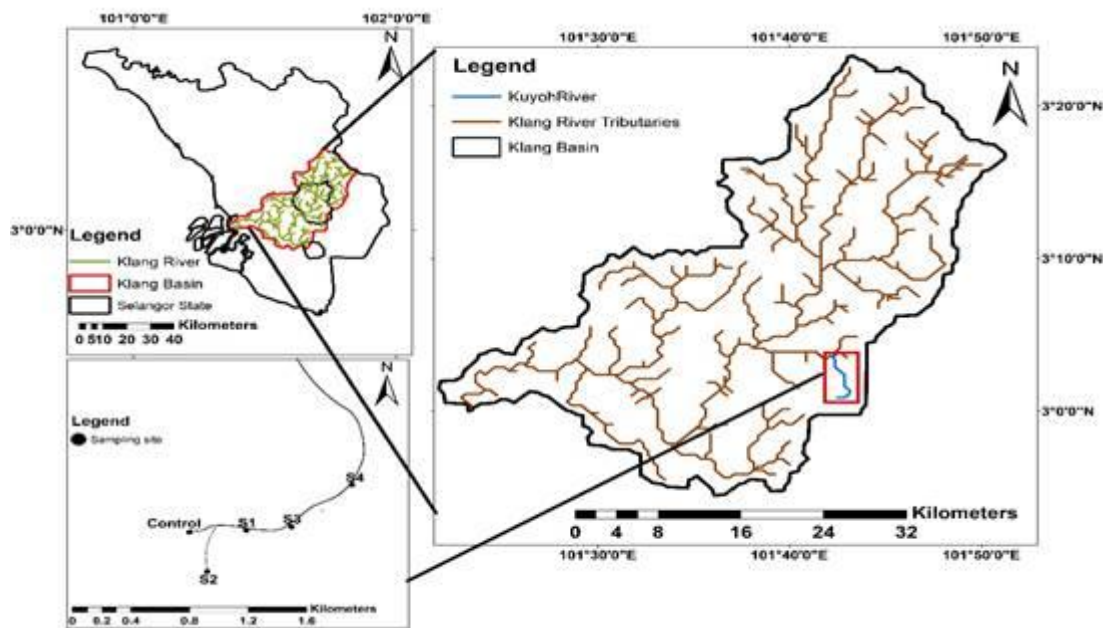
Three questions guide the study: Does contamination follow index-predicted spatial patterns? Does macroinvertebrate composition respond to a metal gradient separable from organic enrichment? Which elements drive this, matching index-dominant ones? We hypothesize organic enrichment accounts for most assemblage variance, but Cd and Cu define a secondary, orthogonal gradient establishing ecological relevance beyond chemical detectability.

The Kuyoh River, a Klang River tributary draining 83 km² of mixed land use in Selangor, Malaysia, spans a gradient from reference reaches to impacted residential, commercial, and industrial stations. Prior studies modeled water quality without trace element speciation or biological data (Hashim & Teo, 2014); this study positions the Kuyoh River as a model for evaluating chemical and biological response as independent evidence in multi-stressor catchments.

2. Materials and Methods

2.1 Study Area and Sampling Design

The Kuyoh River, a 12.5 km Klang River tributary, drains an 83 km² catchment in Selangor and Kuala Lumpur, Malaysia. Five stations were established along an upstream-to-downstream gradient: Control (Lat 3.00542°N, Long 101.710185°E), minimally disturbed, and S1–S4 representing residential, commercial, industrial/sewage, and mixed impacts (Table 1). Monthly sampling occurred during dry (May–August) and wet (November–February) seasons, with three replicate water samples per station per month for physicochemical and trace element analyses. This replicate-and-seasonal design provides robust statistical power for multivariate inference (Hashim & Teo, 2014). The study area, sampling stations, and their surrounding land use are illustrated in Figures 1 and 2



<https://desktop.arcgis.com/> Figure 1: Sample locations in the Kuyoh River, Selangor, Malaysia. (a) Klang Basin location in Selangor State, Peninsular Malaysia; (b) Klang Basin river network showing the Kuyoh River; (c) sampling sites (Control = upstream reference; S1–S2 midstream; S3–S4 downstream). Map created using ArcMap 10.8.2 (Esri Inc., Redlands, CA, USA).



Figure 2: Satellite image (Google Earth, 2025) of the five sampling stations along the Kuyoh River, showing spatial relationship to surrounding residential, commercial, and institutional land uses.

Table 1. Summary of sampling station characteristics.

Station	Coordinates	Land Use
Control	3.00542°N, 101.710185°E	Vegetated, minimal disturbance
S1	3.005191°N, 101.713381°E	Dense residential; restaurant discharges
S2	3.009975°N, 101.710210°E	Urban commercial; channel discharges
S3	3.013789°N, 101.715048°E	Industrial hub; sewage and industrial effluents
S4	3.020926°N, 101.708694°E	Residential-commercial; inlet flows

2.2 Physicochemical and Trace Element Analysis

In-situ DO and pH were measured using calibrated portable meters (YSI 52CE; Eutech pH 450). Laboratory analyses followed APHA (2012): TSS (2540D), COD (HACH DR2800 reactor digestion), BOD₅ (5-day, 20°C, 5210B), NH₃-N (Nessler method, HACH DR2800). Six potentially toxic elements (PTEs: Cd, Cu, Zn, Pb, As, Mn) were determined via ICP-MS (ELAN DRC-II) after acidification (HNO₃, pH <2) and 0.45 µm filtration within 24 hours of collection. 'PTEs' replaces 'heavy metals' per current best practice (Sama et al., 2024; Pakoksung et al., 2025).

2.3 Quality Assurance and Quality Control

Instrument calibration used a five-point curve per element ($R^2 \geq 0.999$). Method Detection Limits ($3 \times \text{SD}$ of seven blank replicates) were: Cd 0.005 $\mu\text{g/L}$, Cu 0.02 $\mu\text{g/L}$, Zn 0.05 $\mu\text{g/L}$, Pb 0.01 $\mu\text{g/L}$, As 0.02 $\mu\text{g/L}$, Mn 0.05 $\mu\text{g/L}$. Certified reference material (NIST SRM 1643f) was analyzed in triplicate at each run's start/end; recoveries were 94–106%. Spiked recoveries (50 $\mu\text{g/L}$) were 91–108%. Values below MDL were set at $\frac{1}{2}$ MDL. Cd and Cu, two to three orders of magnitude below 50 $\mu\text{g/L}$, were verified with 0.5 $\mu\text{g/L}$ spikes (recoveries 89–112%), since Cd at S3–S4 contributes to ERI and CCA2 (George et al., 2021). BDL occurred only in wet season (Mazilamani et al., 2025).

2.4 Pollution and Ecological Risk Indices

Three pollution indices were calculated from station-mean PTE concentrations, kept separate from raw-variable correlation/PCA (Section 2.5) to avoid circularity. The Heavy Metal Pollution Index (Mohan et al., 2024) was computed as $\text{HPI} = \Sigma(W_i Q_i) / \Sigma W_i$, with $W_i = k / S_i$ ($k = 1$; S_i = WHO/Malaysian guideline: Cd 3 $\mu\text{g/L}$, Cu 2000 $\mu\text{g/L}$, Zn 3000 $\mu\text{g/L}$, Pb 10 $\mu\text{g/L}$, As 10 $\mu\text{g/L}$, Mn 100 $\mu\text{g/L}$). The Metal Pollution Index MPI (Haque et al., 2022) is the geometric mean of all PTEs per station. The Ecological Risk Index ERI (Hakanson, 1980) is $\text{ERI} = \Sigma E_r^i$, with $E_r^i = T_r^i \times C_f^i$; toxic-response factors T_r : Cd 30, Cu 5, Pb 5, As 10, Zn 1, Mn 1; $C_f^i = C_i(\text{measured}) / C_i(\text{background})$. Background values were mean dry-season PTEs at Control (Cd 0.038, Cu 1.041, Zn 55.21, Pb 2.537, As 4.418, Mn 26.80 $\mu\text{g/L}$), the least-disturbed local baseline. ERI values here are relative, catchment-specific rankings for dissolved metals, not absolute sediment-based classifications; Control's ERI thus carries no information, while other stations express enrichment relative to it.

2.5 Multivariate Statistical Analysis

A critical principle was maintained: raw variables and composite indices were never included in the same analysis, since HPI, ERI, and MPI derive from the same PTE concentrations and simultaneous inclusion would produce tautological correlations ($r \rightarrow 1.000$). Pearson correlations and PCA were thus conducted exclusively on raw variables (six physicochemical parameters, six PTEs) using the full replicate-level dataset. KMO adequacy and Bartlett's test were computed before PCA, with Varimax rotation (eigenvalues > 1). Analyses were performed in PAST v4.03

and R v4.3.1, with constrained ordination and permutation testing in vegan v2.6-4 (Hammer et al., 2001).

2.6 Macroinvertebrate Sampling and Community Analysis

Macroinvertebrates were collected via 500 micron D-frame kick-sampling across riffle, run, and pool microhabitats, five replicates/station, pooled per trip; (Merritt and Cummins, 1997), identified to family level. Taxon richness, EPT richness, Shannon diversity, and Family Biotic Index were tested by Kruskal-Wallis with Dunn's Holm-corrected post-hoc (Gill, 1998; Young, 2004). CCA in PAST v4.03 related family abundance to twelve environmental variables, testing whether PTEs structure assemblages independently of organic pollution, using dry-season data (Cd, Cu, Pb, As were below detection in wet season). Rare-species downweighting and Monte Carlo permutation (999) tested significance; partial CCA isolated trace-element-attributable variance.

3. Results and Discussion

3.1 Spatial Variation in Physicochemical Parameters and Trace Elements

Physicochemical parameters deteriorated significantly and progressively from Control to S3–S4 (Table 2): DO declined from 6.24 ± 0.74 mg/L (Class I) to 1.42 ± 0.90 mg/L (Class IV), while $\text{NH}_3\text{-N}$ rose from 0.06 ± 0.02 mg/L (Class I) to 3.86 ± 1.19 mg/L (Class V). Mean PTE concentrations increased downstream for all six elements (Table 3): manganese (26.80–240.36 $\mu\text{g/L}$) exceeded Class II at S1–S4 and Class III at S3; arsenic exceeded the 10 $\mu\text{g/L}$ drinking-water guideline at S1–S4 (14.65–23.92 $\mu\text{g/L}$). Cd, Cu, Pb, and As were below detection in the wet season; Zn/Mn remained detectable at impacted stations. Kruskal-Wallis or Welch's ANOVA confirmed significant spatial variation for all six parameters, with post-hoc tests identifying pairwise differences.

Table 2. Mean $\pm$ SD physicochemical water quality parameters by station (both seasons combined, n = 24 per station).

Parameter	Control	S1	S2	S3	S4
DO (mg/L)	6.24 ± 0.74^a	3.18 ± 0.58^b	2.07 ± 1.07^c	1.55 ± 0.93^c	1.42 ± 0.90^c
BOD ₅ (mg/L)	0.47 ± 0.27^a	1.36 ± 0.52^b	2.53 ± 0.95^c	3.14 ± 1.00^c	3.98 ± 1.39^d
COD (mg/L)	7.50 ± 1.82^a	20.93 ± 7.06^b	29.56 ± 8.50^c	39.62 ± 9.44^d	47.88 ± 16.10^d
$\text{NH}_3\text{-N}$ (mg/L)	0.06 ± 0.02^a	1.83 ± 0.58^b	2.56 ± 0.64^b	3.23 ± 0.91^c	3.86 ± 1.19^c

TSS (mg/L)	7.90±2.33 ^a	11.75±2.51 ^b	11.44±2.85 ^b	18.00±5.23 ^c	19.12±5.10 ^c
pH	7.00±0.12 ^a	7.18±0.23 ^b	7.21±0.29 ^b	7.20±0.27 ^b	7.19±0.29 ^b

Different superscript letters indicate statistically significant differences between stations ($p < 0.05$). DO, NH₃-N, TSS: Kruskal-Wallis + Dunn's post-hoc (Holm); BOD₅, COD, pH: Welch's ANOVA + Games-Howell.

This downstream pattern of oxygen depletion and NH₃-N enrichment mirrors other Klang River tributaries, where NH₃-N similarly rose from low upstream concentrations to Class IV under urban/industrial pressure (Jesi et al., 2024), indicating organic wastewater and non-point urban discharge drive deterioration rather than a site-specific anomaly, consistent with other rapidly urbanizing Malaysian rivers (Wan Abdul Ghani et al., 2018).

Mean dissolved PTE concentrations (Cd, Cu, Zn, Pb, As, Mn) increased significantly from the Control station toward S3/S4 (Table 3); Kruskal–Wallis H tests confirmed significant spatial variation for every element ($p \leq 0.05$), with Dunn's post-hoc (Bonferroni-adjusted) identifying pairwise differences via compact letter groupings.

Table 3. Mean ± SD dissolved potentially toxic element (PTE) concentrations (µg/L) at each station, dry season (n = 12 per station).

PTE	Control	S1	S2	S3	S4	H (4)	p
Cd	0.038±0.019 ^a	0.059±0.026 ^a	0.142±0.150 ^{ab}	1.292±1.891 ^{bc}	1.402±1.848 ^c	34.04	< 0.001
Cu	1.041±0.107 ^a	1.383±0.231 ^{ab}	1.712±0.447 ^{bc}	5.287±5.617 ^c	4.397±2.394 ^c	45.44	< 0.001
Zn	55.212±12.833 ^a	76.948±16.840 ^{ab}	93.333±11.386 ^{bc}	104.452±10.126 ^c	122.211±47.70 ^c	37.60	< 0.001
Pb	2.537±0.851 ^a	3.255±0.717 ^{ab}	3.694±1.057 ^{ab}	3.947±1.316 ^{ab}	4.253±0.915 ^b	14.86	0.005
As	4.418±0.595 ^a	14.645±1.385 ^{ab}	16.753±1.037 ^{bc}	19.481±2.038 ^{cd}	23.922±3.031 ^d	53.89	< 0.001
Mn	26.804±15.820 ^a	131.231±15.86 ^{ab}	149.917±12.298 ^b	240.359±34.932 ^c	160.491±8.30 ^{bc}	52.43	< 0.001

Kruskal–Wallis + Dunn's post-hoc (Holm correction). Different superscript letters indicate statistically significant differences ($p \leq 0.05$). Cd, Cu, Pb, As were below detection limit (BDL) at all stations in the wet season.

Arsenic and manganese showed the clearest station separation, increasing roughly 5-fold and 9-fold respectively, consistent with redox-sensitive contamination in Southeast Asian rivers (Hoang et al., 2024; Niampradit et al., 2024). Cd and Cu remained comparable through S1–S2, then rose sharply at

S3–S4, indicating localized industrial discharge (Bahri et al., 2024), while Zn and Pb rose gradually, consistent with diffuse runoff and infrastructure corrosion.

3.2 Raw-Variable Correlation Matrices

The Pearson correlation matrix for raw physicochemical variables (Table 4) showed real associations ranging $|r| = 0.09$ – 0.92 , contrasting sharply with station–mean matrices approaching $r = \pm 1.0$. Strong negative associations between DO and organic loading ($r = -0.838$ to -0.916) confirmed the microbial decomposition–oxygen depletion linkage, while pH's weak TSS association ($r = 0.093$, ns) is consistent with its independent secondary PCA gradient.

Table 4. Pearson correlation matrix: raw physicochemical variables.

	DO	BOD ₅	COD	NH ₃ -N	TSS	pH
DO	1.000					
BOD ₅	−0.838***	1.000				
COD	−0.778***	0.762***	1.000			
NH ₃ -N	−0.916***	0.907***	0.804***	1.000		
TSS	−0.559***	0.516***	0.606***	0.609***	1.000	
pH	−0.470***	0.542***	0.247**	0.546***	0.093 ns	1.000

*** $p < 0.001$; ** $p < 0.01$; ns = not significant.

The PTE correlation matrix (Table 5) showed a similarly realistic range ($|r| = 0.04$ – 0.77): Cd showed almost no linear association with Cu ($r = 0.036$, ns), consistent with their separate PC loadings (Section 3.3), while As–Mn ($r = 0.765$) and As–Zn ($r = 0.726$) showed the strongest associations, supporting co-loading on the primary contamination gradient.

Table 5. Pearson correlation matrix: raw PTE variables.

	Cd	Cu	Zn	Pb	As	Mn
Cd	1.000					
Cu	0.036 ns	1.000				
Zn	0.220 ns	0.336**	1.000			

Pb	0.382**	0.361**	0.548***	1.000		
As	0.383**	0.506***	0.726***	0.463***	1.000	
Mn	0.244 ns	0.441***	0.483***	0.341**	0.765***	1.000

*** $p < 0.001$; ** $p < 0.01$; ns = not significant.

Moderate correlations are consistent with heterogeneous associations elsewhere (Dutta et al., 2018). Analyses based on few spatial units can inflate relationships when nested observations are treated as independent, constituting pseudoreplication (Hurlbert, 1984); using independent replicate-level observations instead preserves degrees of freedom and better represents genuine covariation.

3.3 Principal Component Analysis of Raw Variables

PCA was suitable ($KMO = 0.844$; Bartlett's $\chi^2(15) = 677.75$, $p < 0.001$), yielding two components with eigenvalues > 1 , explaining 86.2% variance (Table 6). PC1 (69.9%) reflected organic enrichment/oxygen depletion through NH_3-N (0.977), DO (-0.937), BOD_5 (0.934), COD (0.869), and TSS (0.682), while PC2 (16.3%) captured an independent acid-base dimension defined by pH (0.776) and TSS (-0.544).

Table 6. PCA results for raw physicochemical variables (varimax-rotated loadings).

Variable	PC1	PC2
DO	-0.937	-0.015
BOD_5	0.934	0.118
COD	0.869	-0.266
NH_3-N	0.977	0.052
TSS	0.682	-0.544
pH	0.550	0.776
Eigenvalue	4.229	0.986
Variance explained (%)	69.9	16.3
Cumulative (%)	69.9	86.2

$KMO = 0.844$; Bartlett's $\chi^2(15) = 677.75$, $p < 0.001$.

PTE PCA was adequate ($KMO = 0.656$; Bartlett's $\chi^2(15) = 153.36$, $p < 0.001$), with two components explaining 69.8% variance (Table 7). PC1 (53.0%) reflected mixed-source

contamination through As (0.923), Mn (0.792), Zn (0.804), Cu (0.624), and Pb (0.707), while PC2 (16.8%) contrasted Cd (0.792) with Cu (−0.518), indicating localized inputs — the lower cumulative variance (versus 87–98% for station-mean PCA) reflecting genuine sampling adequacy, not model weakness.

Table 7. PCA results for raw PTE variables (varimax-rotated loadings).

PTE	PC1	PC2
Cd	0.467	0.792
Cu	0.624	−0.518
Zn	0.804	−0.015
Pb	0.707	0.281
As	0.923	−0.072
Mn	0.792	−0.210
Eigenvalue	3.233	1.025
Variance explained (%)	53.0	16.8
Cumulative (%)	53.0	69.8

$KMO = 0.656$; $Bartlett's \chi^2(15) = 153.36, p < 0.001$.

Separating raw-variable analysis from index reporting improved validity (Hurlbert, 1984): PTE correlations were plausible ($r = 0.04$ – 0.77), unlike $r > 0.99$ from station means with $n = 5$ (Schönbrodt & Perugini, 2013), and the 69.8% variance solution, below the 87–98% typical of station-mean PCA, avoids high-dimensional, low-sample instability, supporting reproducible source apportionment (Björklund, 2019).

3.4 Pollution and Ecological Risk Indices

HPI, ERI, and MPI increased from Control to S3/S4 (Table 8): ERI reached 1099 at S3 and 1189 at S4, exceeding the very high-risk threshold of 600 (Hakanson, 1980), largely due to Cd's $T_r = 30$. HPI stayed below 100 (2.11 at Control, 9.58 at S3, 9.06 at S4; Mohan et al., 2024), and MPI rose more than fivefold (2.95 to 15.4, 15.2). ERI thus indicated very high risk, MPI cumulative loading, and HPI acceptable water, a divergence reflecting index weighting: HPI dilutes Mn/As exceedances at S1–S4, whereas ERI is Cd-dominated.

Table 8. Pollution and ecological risk indices by station (dry season). All indices computed from station-mean PTE concentrations, kept analytically separate from raw-variable PCA.

Station	HPI	MPI	ERI	ERI classification
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Control	1.83	2.95	52.0	Low risk (<150)
S1	5.79	5.83	98.8	Low risk (<150)
S2	6.66	7.72	172.2	Moderate risk (150–300)
S3	9.58	15.37	1099.0	Very high risk (>600)
S4	9.06	15.19	1188.8	Very high risk (>600)

ERI classification thresholds follow Hakanson (1980): <150 low risk, 150–300 moderate, 300–600 considerable, >600 very high. Background values for ERI: Control-station mean dry-season concentrations (Section 2.4).

This disagreement is not unique: Edet & Offiong (2002) found two heavy-metal indices applied to the same Nigerian samples produced contradictory classifications, prompting a third index. Its persistence for two decades shows no single index should define severity without cross-validation against alternatives or, preferably, biological data.

3.5 Macroinvertebrate Assemblage Structure

Assemblage conditions declined with deterioration, though responses varied: taxon richness decreased from Control to S4, while EPT taxa, present at Control and S1, were absent from S3/S4 throughout. Impacted stations were dominated by *Oligochaetes* (*Tubificidae*, *Naididae*, *Lumbriculidae*), tolerant of low DO and organic enrichment, so abundance remained high despite low richness/diversity. Shannon diversity and FBI tracked richness, with all three metrics differing significantly (Kruskal-Wallis, $p < 0.05$; Dunn's Holm post-hoc), Control distinct from S3/S4. This pattern matches responses to organic enrichment (Eriksen et al., 2021) and downstream biotic decline in the Penchala River (Wan Abdul Ghani et al., 2018).

3.6 Biological Validation: CCA of Macroinvertebrate Assemblages Against Combined Environmental Variables

CCA confirmed organic enrichment as the dominant assemblage gradient (CCA1: 89.3%; Table 9), driven by DO (−1.658), NH_3 −N (0.739), and BOD_5 (0.585), mirroring physicochemical PC1, while Cd (−3.492) and Cu (−0.804) loaded strongly on CCA2 (9.2%) and As, Zn, Mn, Pb loaded weakly showing the Cd–Cu factor from PTE PCA (PC2, Section 3.3) was biologically active with independently detectable effects. Environmental variable loadings on the first two CCA axes are detailed in Table 10.

Table 9. CCA eigenvalues and variance explained: combined twelve-variable model.

CCA Axis	Eigenvalue	Variance Explained (%)	Cumulative (%)
Axis 1	0.2935	89.29	89.29
Axis 2	0.0303	9.22	98.51
Axis 3	0.0034	1.04	99.55
Axis 4	0.0015	0.45	100.00

Table 10. CCA scores of environmental variables on Axes 1 and 2 (combined model).

Variable	Axis 1	Axis 2
DO	−1.6581	0.5530
BOD ₅	0.5848	−0.7588
COD	0.5070	−0.5475
NH ₃ -N	0.7389	−0.2257
TSS	0.1362	−0.3765
pH	−0.3557	0.4878
Cd	0.8731	−3.4920
Cu	0.1612	−0.8039
Zn	−0.2481	0.4693
Pb	−0.1960	0.3711
As	0.0312	0.5078
Mn	−0.0397	0.5183

The extreme Cd CCA2 score (−3.492) indicates Cd functions as an independent ecological filter orthogonal to the organic pollution gradient.

Three findings support a genuine ecological signal: Axis 2 significance ($p = 0.004$), VIFs < 10 , and a partial CCA controlling six physicochemical covariables ($p = 0.006$) (ter Braak & te Beest, 2022). Despite right-skewed Cd at S3/S4 (1.292 ± 1.891 ; 1.402 ± 1.848 $\mu\text{g/L}$), CCA2 (9.2%) identified Cd–Cu effects independent of CCA1 organic pollution (89.3%). HPI/ERI/MPI (Table 8) quantify station-level contamination but cannot resolve distinct biological pathways among the six PTEs; constrained ordination thus strengthens risk assessment by linking contaminants directly to assemblage composition (Peng et al., 2021; Peng et al., 2020).

Cd and Cu's independent CCA2 position is mechanistically plausible: Cd at S3–S4 (1.29–1.40 µg/L) can interact with calcium-related uptake pathways and induce oxidative stress (Cuypers et al., 2010; Järup, 2003; Peeters et al., 2000; Poteat & Buchwalter, 2014), while Cu at S3 (4.4–5.3 µg/L) can impair respiratory and neurological function independent of organic oxygen demand (Clements et al., 2000; O’Gara et al., 2004; Wang et al., 2023). Their near-zero correlation ($r = 0.036$; Table 5), despite shared PC2/CCA2 loadings, suggests a localized S3–S4 source distinct from diffuse As/Mn/Zn/Pb inputs, consistent with other effluent-impacted rivers (Arimoro et al., 2021); management should combine catchment controls with source tracing upstream of S3.

PTE patterns matched Asian urban rivers: Hanoi Mn (23.9–307 µg/L) and As near/exceeding limits (Hoang et al., 2024) paralleled Mn's PC1 dominance (0.792) and the S3 peak, consistent with redox-mediated Mn release. Bahri et al. (2024) reported higher Tengi River Zn/Pb (As avg. 10.87 µg/L); Niampradit et al. (2024) likewise found Mn predominant in the Chao Phraya. Yet ERI underweights Mn ($T_r = 1$) versus Cd ($T_r = 30$) (Hakanson, 1980): ERI classified S3/S4 very high risk, HPI rated all stations acceptable, and CCA identified a significant Cd–Cu gradient, with Mn/As breaching NWQS/drinking-water guidelines while Cd/Cu stayed compliant despite driving ERI and assemblage responses.

Sediment-dwelling Tubificidae, Naididae, and Lumbriculidae experience exposure via porewater, sediment ingestion, and dermal contact; sediment-phase measurements should complement dissolved PTEs to test whether the CCA2 Cd-Cu signal persists without seasonal dilution (Biedunkova et al., 2025; Dalu et al., 2017). This framework of raw-variable separation, replicate-level analysis, and biological validation improves inference beyond station-mean indices.

3.7 Limitations

Several limitations constrain interpretation. Only dissolved trace elements were measured; since sediment-dwelling oligochaetes are exposed mainly through porewater and particle-bound metals, total exposure is underestimated, making the dissolved Cd-Cu signal conservative. Family-level identification may mask genus- or species-level losses. Cd-Cu separation relied on dry-season data, since Cd, Cu, Pb, and As were below detection in the wet season, leaving wet-season persistence untested. ERI used dissolved concentrations with sediment-derived toxic-response factors, interpreted only as a relative within-catchment ranking. Finally, one catchment and one annual cycle limit generalisation, requiring confirmation across other tropical urban rivers.

4. Conclusion

Pollution indices alone provide a misleading picture of ecological risk, since they cannot test whether contamination is biologically detectable. Integrating raw-variable PCA with CCA-based validation reveals cadmium and copper structure the Kuyoh River's macroinvertebrate assemblage independently of the organic pollution gradient. The three approaches disagreed: HPI rated all stations acceptable, ERI flagged downstream stations very high risk, while biology implicated Cd and Cu, neither exceeding regulatory thresholds. Manganese, the only threshold-breaching element, carries negligible ERI weighting and remains invisible to it. This framework raw-variable separation, adequately powered analysis, justified background values, and biological validation offers a transferable template for trace element assessment.

Use of AI tools declaration

The authors used editGPT and Grammarly for language clarity.

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Author Contributions EFA: Conceptualization, Methodology, Data curation, Writing -Original draft preparation. NH: Conceptualization, Writing -Reviewing, Editing and Supervision. FMY: Conceptualization, Methodology, Writing -Reviewing, Editing. WLWJ: Conceptualization, Methodology, Investigation, Writing -Reviewing and Editing. AOS: Editing and proofreading, Analysis, R software, EAB: Resources, visualization, data curation and analysis. EMU: Reviewing, software, Editing and Project administration.

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